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высшего образования
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(ФГАОУ ВО «МГТУ «СТАНКИН»)

Институт цифровых
интеллектуальных систем

Кафедра
компьютерных систем управления

Дисциплина «Основы системного программного обеспечения»

Отчет по лабораторной работе №1

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Содержание

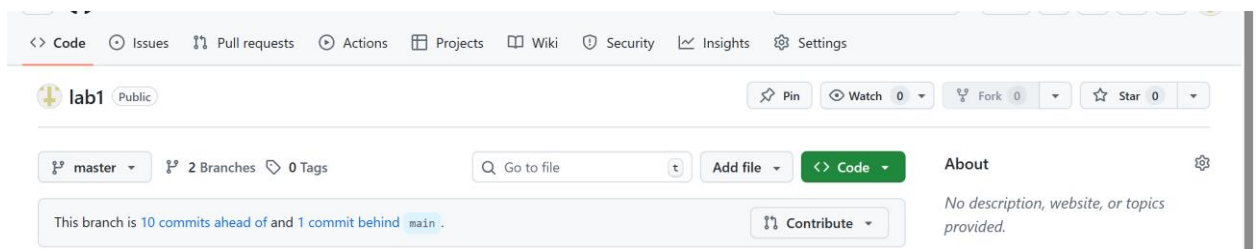
1. Цель: Освоить работу с системой контроля версий Git и платформой GitHub, включая клонирование репозитория, совместную работу, отправку изменений и разрешение конфликтов.

2. Задачи работы:

1. Создать и настроить локальный репозиторий Git.
2. Освоить основные команды Git: add, commit, status, push, pull.
3. Научиться создавать и объединять ветки репозитория.
4. Освоить работу с удалённым репозиторием на GitHub.
5. Настроить файл .gitignore для игнорирования файлов.
6. Выполнить клонирование репозитория с помощью команды git clone.
7. Освоить совместную работу над проектом с использованием GitHub.
8. Выполнить добавление, изменение и отправку файлов в удалённый репозиторий.
9. Получить и разрешить конфликт при совместной работе с репозиторием

3. Описание и выполнение тестовых (если присутствуют) и индивидуальных заданий

1) Создание локального репозитория



```
Командная строка
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Alexey.txt

nothing added to commit but untracked files present (use "git add" to track)
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git remote add origin https://github.com/zeeez-888/lab1
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git remote -v
origin https://github.com/zeeez-888/lab1 (fetch)
origin https://github.com/zeeez-888/lab1 (push)
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git pull origin master
fatal: unable to access 'https://github.com/zeeez-888/lab1/': SSL certificate problem: self-signed certificate in certificate chain
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git push origin master
info: please complete authentication in your browser...
C:\Users\student.STANKIN\Downloads\Чен_наб_1>
C:\Users\student.STANKIN\Downloads\Чен_наб_1>
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git push origin master
C:\Users\student.STANKIN\Downloads\Чен_наб_1>
```

```
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git config --global user.name "Alexey"
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git config --global user.email alexchen417@gmail.com
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git init
Initialized empty Git repository in C:/Users/student.STANKIN/Downloads/Чен_наб_1/.git/
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Alexey.txt
    test.txt.txt
```

```
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git add test.txt.txt
C:\Users\student.STANKIN\Downloads\Чен_наб_1>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   test.txt.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Alexey.txt
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit -m "first commit"
[master (root-commit) 408fb3b] first commit
1 file changed, 1 insertion(+)
create mode 100644 test.txt.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Alexey.txt

nothing added to commit but untracked files present (use "git add" to track)
```

2) Отправка коминат на сервер

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git remote add origin https://github.com/zeeez-888/lab1

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git remote -v
origin https://github.com/zeeez-888/lab1 (fetch)
origin https://github.com/zeeez-888/lab1 (push)
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git log
commit 408fb3bfb391fe9f07fa91ff62590bf667267bf7 (HEAD -> master, origin/master)
Author: Alexey <alexchen417@gmail.com>
Date: Thu Feb 26 13:21:23 2026 +0300

    first commit
```

The screenshot shows the GitHub repository interface for 'zeeez-888/lab1'. At the top, it indicates the current branch is 'master', which is 1 commit ahead of and 1 commit behind the 'main' branch. Below this, a commit history table shows a single commit by 'zeeez-888' titled 'first commit' with hash '408fb3b' and a timestamp of '44 minutes ago'. A file 'test.txt.txt' is listed as part of this commit.

The screenshot displays the details of commit '408fb3b' by 'zeeez-888'. It shows the commit message 'first commit' and a list of changed files, including 'test.txt.txt'. The diff view for 'test.txt.txt' shows a single line change: '1 -> Chen'. A 'Comments' section is visible at the bottom. A 'Customizable line height' notification is present on the right side of the page.

3) Создание новой ветки

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git branch second

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git branch
* master
  second

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git checkout second
Switched to branch 'second'

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>_
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add Chen.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git status
On branch second
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Chen.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>_
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin second
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'second' on GitHub by visiting:
remote:   https://github.com/zeeez-888/lab1/pull/new/second
remote:
To https://github.com/zeeez-888/lab1
 * [new branch]      second -> second

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>_
```

 second ▾

 3 Branches

 0 Tags

Q Go to file 

Add file ▾

 Code ▾

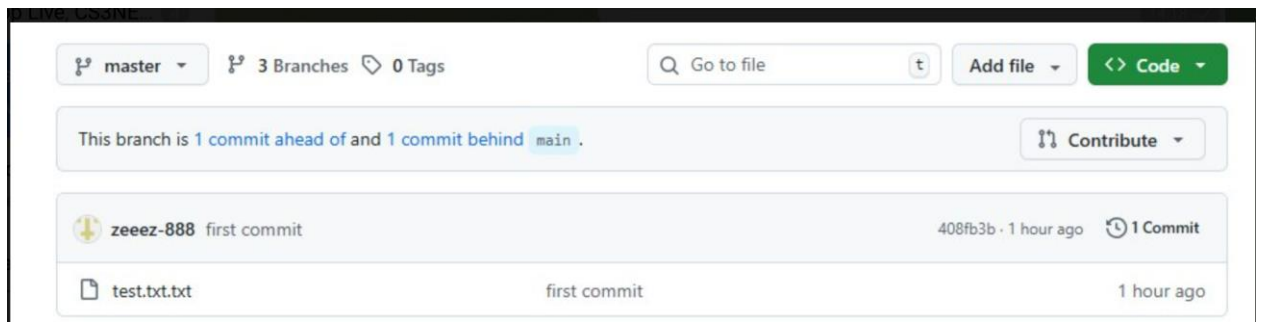
This branch is [2 commits ahead of](#) and [1 commit behind](#) [main](#).

 Contribute ▾

 zeeez-888 second commit 176a6cd · now  2 Commits

 Chen.txt second commit now

 test.txt.txt first commit 1 hour ago



4) Слияние веток

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git checkout master
Switched to branch 'master'
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git merge second
Updating 408fb3b..176a6cd
Fast-forward
 Chen.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 Chen.txt
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git branch -d second
Deleted branch second (was 176a6cd).
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git branch newdev
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git checkout newdev
Switched to branch 'newdev'
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add otchestvo.txt
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git status
```

```
On branch newdev
```

```
Changes to be committed:
```

```
(use "git restore --staged <file>..." to unstage)
    new file:   otchestvo.txt
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit -m "third commit"
```

```
[newdev 78b0d9f] third commit
```

```
1 file changed, 1 insertion(+)
```

```
create mode 100644 otchestvo.txt
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin newdev
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 315 bytes | 315.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'newdev' on GitHub by visiting:
remote:   https://github.com/zeeez-888/lab1/pull/new/newdev
remote:
To https://github.com/zeeez-888/lab1
 * [new branch]      newdev -> newdev
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git checkout master
Switched to branch 'master'
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git merge newdev
Updating 176a6cd..78b0d9f
Fast-forward
 otchestvo.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 otchestvo.txt
```

```
C:\Users\Andrey\OneDrive\Desktop\inzilya>git checkout newdev
Switched to branch 'newdev'
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git checkout newdev
Switched to branch 'newdev'

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add otchestvo.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git status
On branch newdev
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   otchestvo.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin newdev
Everything up-to-date

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit -m "4 commit"
[newdev 43b56d3] 4 commit
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin newdev
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 305 bytes | 76.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/zeeez-888/lab1
 78b0d9f..43b56d3  newdev -> newdev

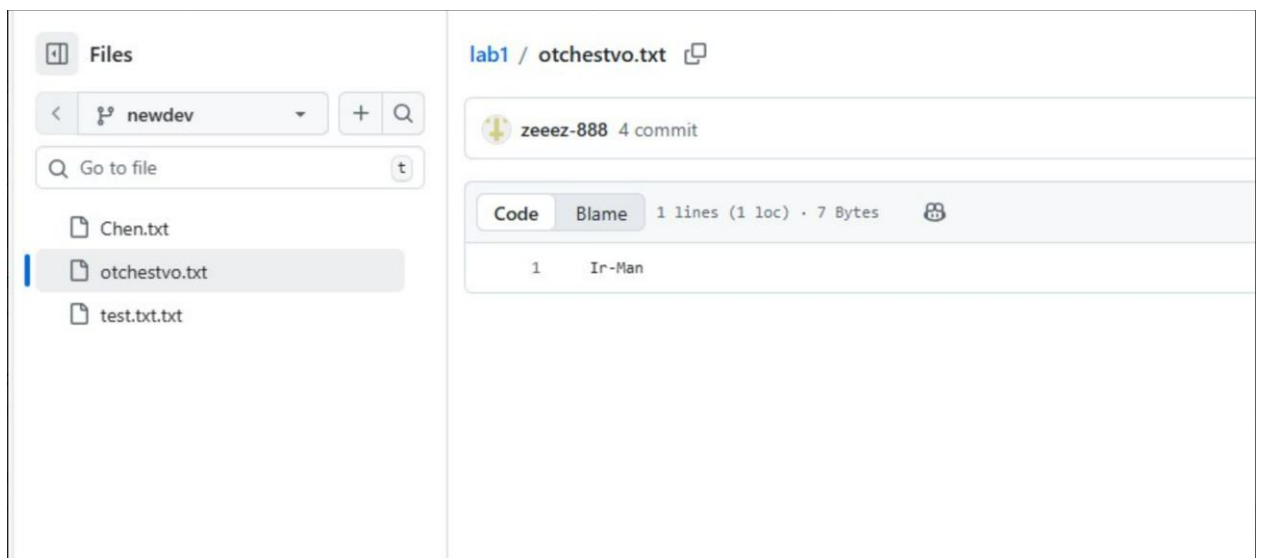
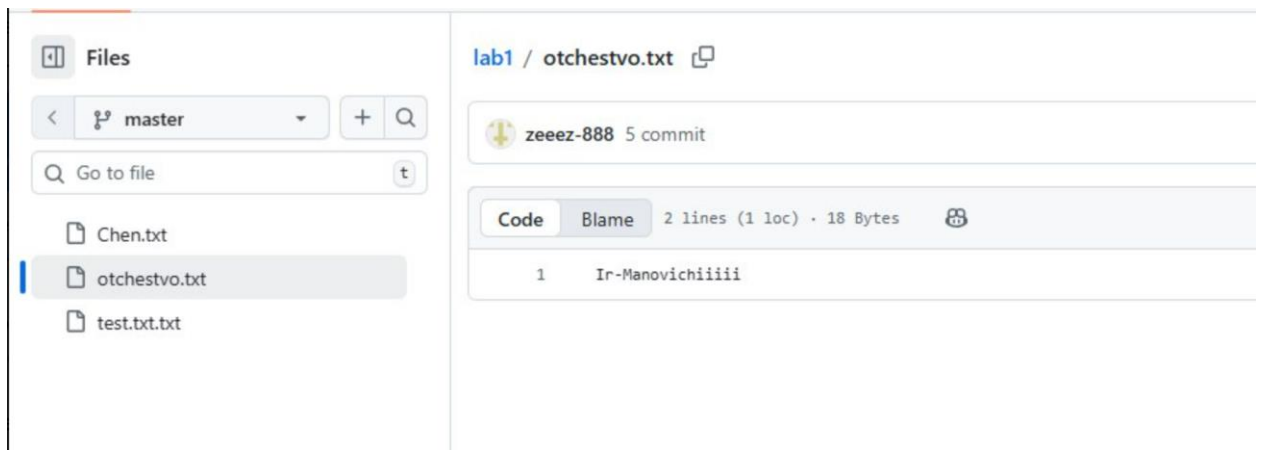
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>_
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git checkout master
Switched to branch 'master'

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add otchestvo.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit -m "5 commit"
[master b99b531] 5 commit
 1 file changed, 2 insertions(+), 1 deletion(-)

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 316 bytes | 316.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/zeeez-888/lab1
 408fb3b..b99b531  master -> master
```



```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git merge newdev
Auto-merging otchestvo.txt
CONFLICT (content): Merge conflict in otchestvo.txt
Automatic merge failed; fix conflicts and then commit the result.
```

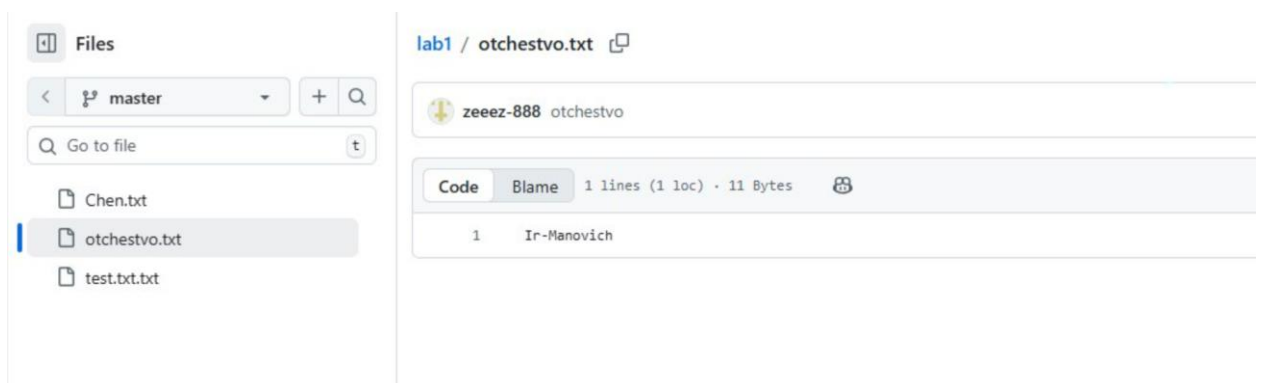
5) Просмотр изменений и разрешение конфликтов

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git diff
diff --cc otchestvo.txt
index abaf1dd,3cafaf7..0000000
--- a/otchestvo.txt
+++ b/otchestvo.txt
@@@ -1,2 -1,1 +1,6 @@@
++<<<<<< HEAD
+Ir-Manovichiiii
+
++=====
+ Ir-Man
++>>>>>> newdev
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add otchestvo.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit -m "otchestvo"
[master ebfecea] otchestvo

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin master
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 341 bytes | 341.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/zeeez-888/lab1
   b99b531..ebfecea  master -> master
```



The image shows a side-by-side comparison of a local file explorer and a GitHub repository view. On the left, a file explorer window titled 'Files' shows the contents of a directory named 'lab1'. It lists three files: 'Chen.txt', 'otchestvo.txt' (which is selected), and 'test.txt.txt'. On the right, the GitHub web interface shows the 'otchestvo.txt' file in the 'lab1' repository. The commit is by user 'zeeez-888' with the message 'otchestvo'. The file content is displayed as 'Ir-Manovich'.

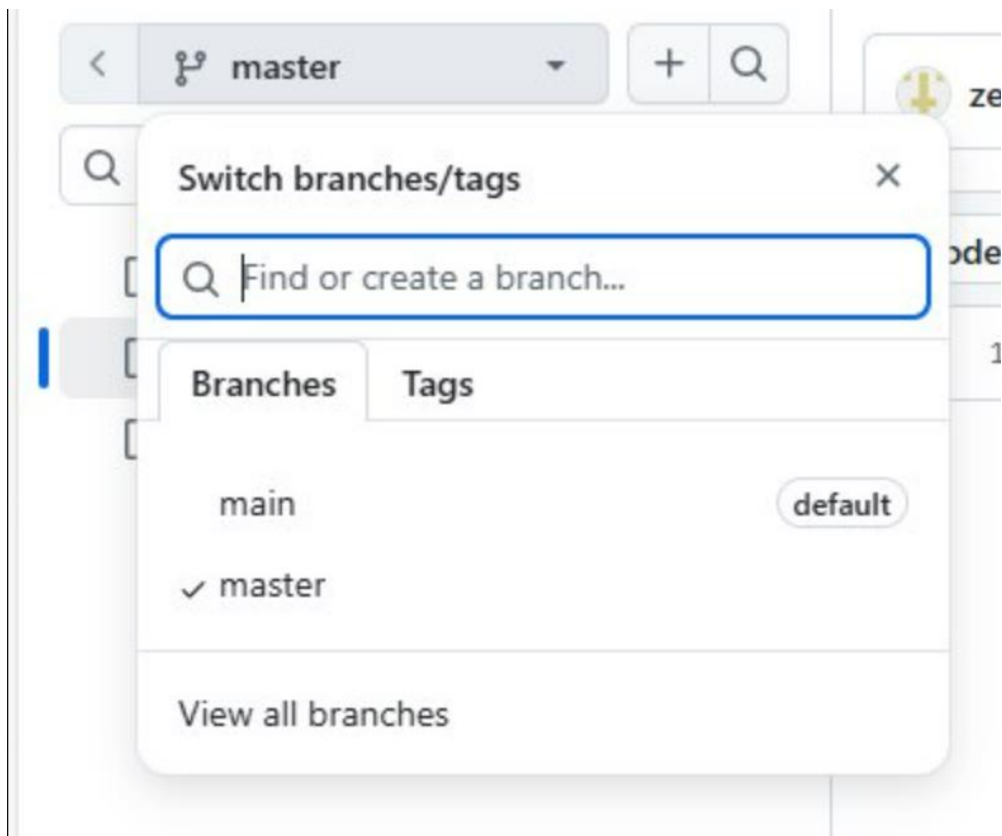
6) Удаление веток на сервере

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git branch -d newdev
Deleted branch newdev (was 43b56d3).
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin --delete newdev
To https://github.com/zeeez-888/lab1
- [deleted]          newdev

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin --delete newdevsecond
error: unable to delete 'newdevsecond': remote ref does not exist
error: failed to push some refs to 'https://github.com/zeeez-888/lab1'

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin --delete second
To https://github.com/zeeez-888/lab1
- [deleted]          second
```



7) Возврат к предыдущему состоянию

```

C:\Users\student.STANKIN\Downloads\Чен_наб_1>git log
commit ebfecea81eef2559e8f776cbb8555a089bcd250a (HEAD -> master, origin/master)
Merge: b99b531 43b56d3
Author: Alexey <alexchen417@gmail.com>
Date: Thu Feb 26 14:50:59 2026 +0300

    otchestvo

commit b99b531f0d2976ee642f4a2bab869bcab026e5af
Author: Alexey <alexchen417@gmail.com>
Date: Thu Feb 26 14:44:56 2026 +0300

    5 commit

commit 43b56d31815e58a42e3ca19b2d62e7d78046098b
Author: Alexey <alexchen417@gmail.com>
Date: Thu Feb 26 14:43:04 2026 +0300

    4 commit

commit 78b0d9f01778a9c28144f0a6e0a8ff66270ac418
Author: Alexey <alexchen417@gmail.com>
Date: Thu Feb 26 14:26:38 2026 +0300

    third commit

commit 176a6cd476a7b72decee665b4ef1e3861d83c19c
Author: Alexey <alexchen417@gmail.com>
Date: Thu Feb 26 14:17:54 2026 +0300

    second commit

commit 408fb3bf391fe9f07fa91ff62590bf667267bf7
Author: Alexey <alexchen417@gmail.com>
Date: Thu Feb 26 13:21:23 2026 +0300

    first commit

```

Commits

master	All users	All time
Commits on Feb 26, 2026		
otchestvo	zeez-888 committed 6 minutes ago	ebfecea <>
5 commit	zeez-888 committed 12 minutes ago	b99b531 <>
4 commit	zeez-888 committed 14 minutes ago	43b56d3 <>
third commit	zeez-888 committed 30 minutes ago	78b0d9f <>
second commit	zeez-888 committed 39 minutes ago	176a6cd <>
first commit	zeez-888 committed 1 hour ago	408fb3b <>

1 file changed +1 -0 lines changed

Chen.txt

```

*** @@ -0,0 +1 @@
1 + Chen

```

Comments

Lock conversation

=

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git checkout 408fb3bfb391fe9f07fa91ff62590bf667267bf7
Note: switching to '408fb3bfb391fe9f07fa91ff62590bf667267bf7'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

    git switch -c <new-branch-name>

Or undo this operation with:

    git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 408fb3b first commit
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git branch
* (HEAD detached at 408fb3b)
  master
```

8) Исправление коммита

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git checkout master
Previous HEAD position was 408fb3b first commit
Switched to branch 'master'

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add test.txt.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   test.txt.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git reset head
Unstaged changes after reset:
M      test.txt.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test.txt.txt

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add test.txt.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit -m "6 commit"
[master 08257f4] 6 commit
 1 file changed, 1 insertion(+), 1 deletion(-)

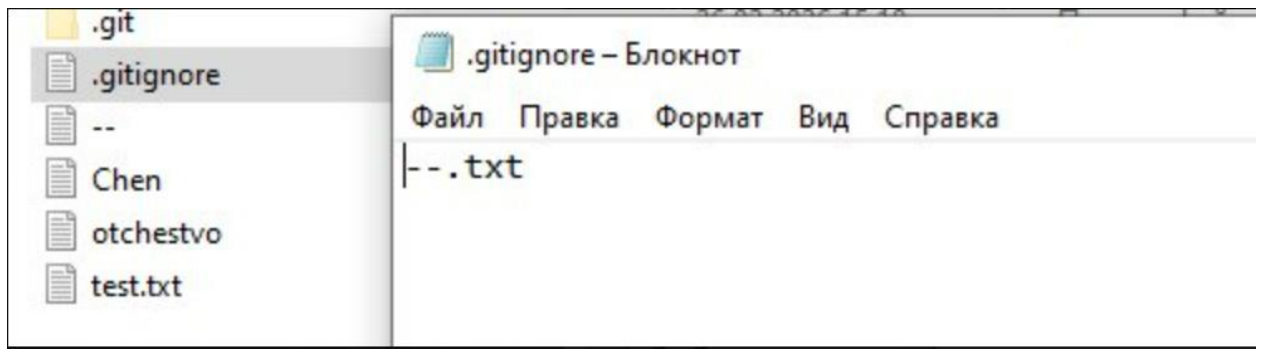
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git reset head

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git reset head

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git status
On branch master
nothing to commit, working tree clean
```

9) Отправка только нужных файлов на сервер

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit --amend
[master 259c452] 6 commit
Date: Thu Feb 26 15:05:46 2026 +0300
 1 file changed, 1 insertion(+), 1 deletion(-)
```



```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add .gitignore.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit -m "7 commit ignore"
[master 6272b6b] 7 commit ignore
1 file changed, 1 insertion(+)
create mode 100644 .gitignore.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin master
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 539 bytes | 539.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/zeeez-888/lab1
ebfecea..6272b6b master -> master
```

```
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git add .gitignore.txt

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git commit -m "8 commit ignore"
[master ed9ee86] 8 commit ignore
1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\student.STANKIN\Downloads\Чен_лаб_1>
C:\Users\student.STANKIN\Downloads\Чен_лаб_1>git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 263 bytes | 263.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/zeeez-888/lab1
6272b6b..ed9ee86 master -> master
```

